

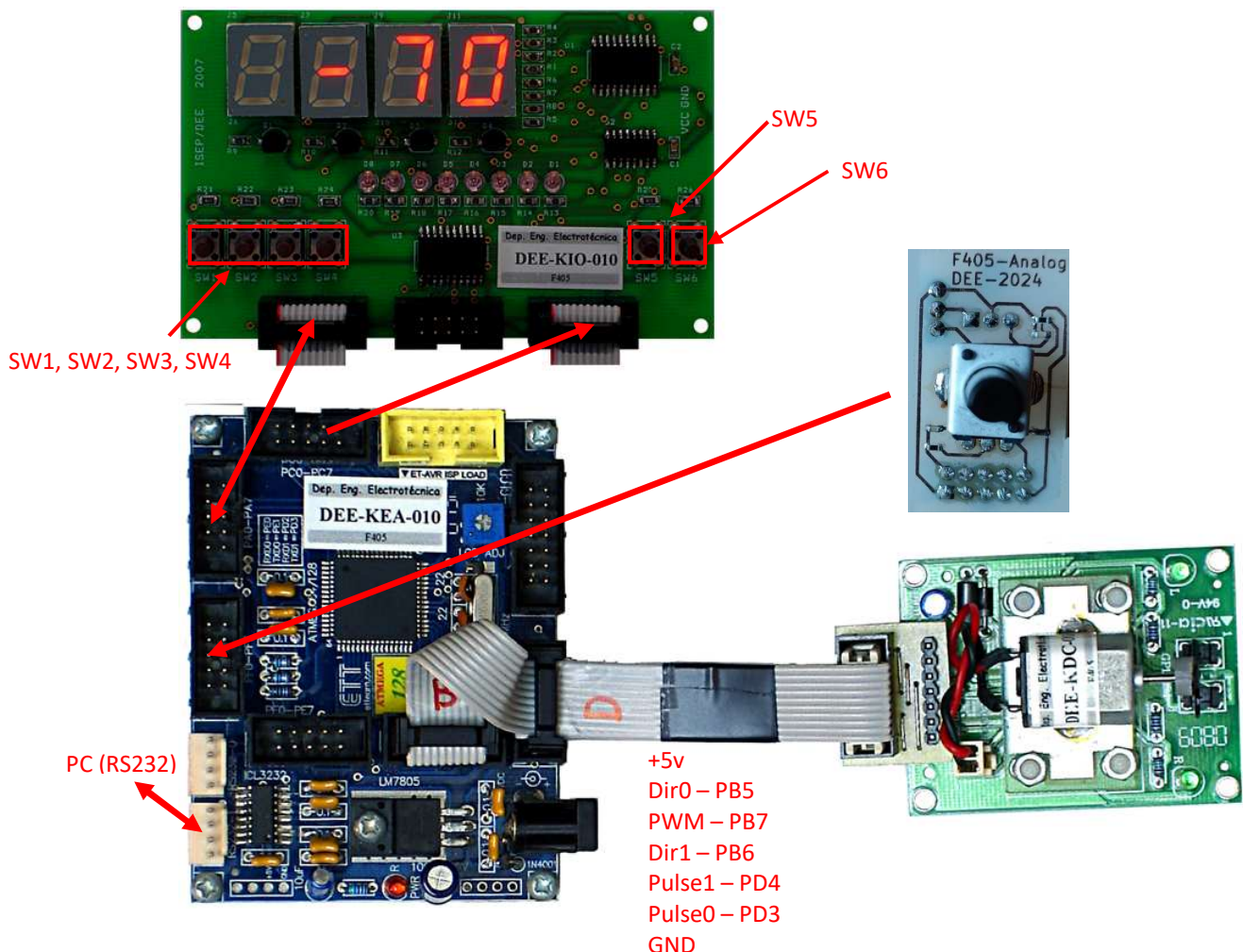
Project 2

Objective: Programming using C, open-loop speed control of a DC motor, asynchronous serial communication, A/D converter, stepper motor control (MPP) and encoder.

Requirements: Knowledge of C programming language, timer counters, interrupts, USART and A/D converter.

Operation 1: The objective is to control the speed of a DC motor using PWM. The switches **SW1**, **SW2** increment (**SW1**) and decrement (**SW2**) the percentage value of the nominal motor speed by 5 units. The switches **SW3** and **SW4** define the nominal motor speed with values 25% and 50%, respectively. The percentage value of the nominal motor speed must be shown in the display. Every time the speed is changed, it's value must be updated in the display. Switch **SW5** reverses the direction of the motor's rotation and **SW6** should stop the motor.

Necessary hardware:



Operation specifics:

The percentage value of the motor's rotation speed is shown in the display. When the motor's rotation direction is counterclockwise, a sign (-) should be shown in *display 2*. To reverse the motor's rotation direction, the motor should be previously stopped for a period of **250 ms**.

Suggestion:

- Use the **TC0**, in **CTC** mode, to obtain a time base of **5 ms**.
- Use the **TC2**, in **PWM** mode (**Fast PWM** or **Phase Correct PWM**), to generate the PWM signal in the **OC2** output (PB7). The frequency of the PWM signal should be, approximately, **500 Hz**. The rotation direction is defined by the outputs **Dir0** (PB5) and **Dir1** (PB6).

Software implementation:

Use **C programming language**.

Operation 2:

Using **USART1** for the asynchronous serial communication (RS232), with **9600 bps**, **8** bits of data and **2** stop bits, the objective is to control the motor's speed using the PC. The characters to be sent to the ATmega 128 are:

Character	Description
"P" or "p"	Stops the motor
"1"	25% of nominal speed (direction defined by SW5)
"2"	50% of nominal speed (direction defined by SW5)
"+"	Increments 5% of the nominal speed
"_"	Decrement 5% of the nominal speed
"I" or "i"	Reverses the rotation direction
"B" or "b"	Request to send the current duty cycle of the motor (in percentage)

To select the mode of operation of the system, two new characters need to be added. The character "S" selects the *Switches* operating mode (controlled by the switches) and the character "D" selects the Digital operating mode (controlled by the PC).

The system must start with the Digital operating mode selected. The display 3 should indicate the current operating mode at all times.

Software implementation:

Use **C programming language**.

Operation 3:

The objective is to control the nominal motor speed using the voltage value obtained at the terminals of a potentiometer. The voltage value ranges from 0V to 5V and the motor's speed must change from 0 RPM (0V) to the nominal motor speed (5V). The value of the percentage of the motor speed must be shown in the display. Switch **SW5** reverses the direction of the motor's rotation.

A new operating mode must be added (Analog mode "A").

Character	Description
"S" or "s"	Switches Mode
"D" or "d"	Digital Mode
"A" or "a"	Analog Mode

Analog data acquisition:

The analog/digital conversion of the voltage given by the potentiometer should be obtained by calculating the average value of 4 consecutive readings of the A/D converter (**ADC0**), using an accuracy of **8 bits**.

The data acquisition routine should be programmed in Assembly.

Software implementation:

Use **C programming language** + **Assembly programming language** (data acquisition routine)

Operation 4:

The objective is to control the position of a stepper motor (20 steps/rotation or 40 half steps/rotation) using the **USART1**. The control of the stepper motor's position is done by sending a command with the number of steps (to reach the desired motor position) and ending with the "Enter" keyboard. The frame received by the **USART1** must have the following structure:

Value	\r	\n
-------	----	----

The following characters must be considered:

Character	Description
"R" or "r"	Defines a reference position
"+"	Increments 1 step
"_"	Decrements 1 step

The time between two motor steps must be, approximately, **25 ms**. The display must show the current number of steps from the reference position.

A new operating mode must be added (Stepper motor mode "M")

Character	Description
“S” or “s”	Switches Mode
“D” or “d”	Digital Mode
“A” or “a”	Analog Mode
“M” or “m”	Stepper Motor Mode

Tables of the stepper motor drive modes:

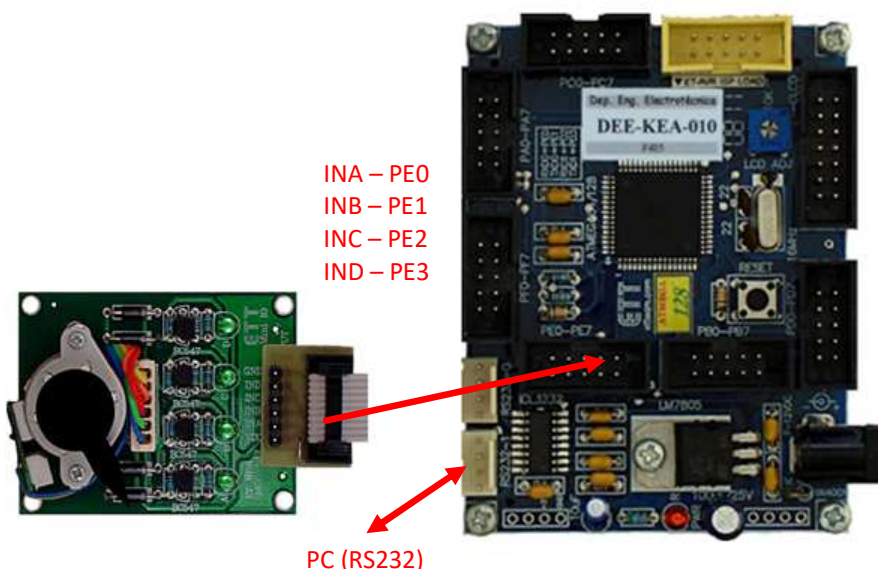
1. Full step drive

IND	INC	INB	INA
1	0	0	1
1	1	0	0
0	1	1	0
0	0	1	1

• Half step drive

IND	INC	INB	INA
0	0	0	1
1	0	0	1
1	0	0	0
1	1	0	0
0	1	0	0
0	1	1	0
0	0	1	0
0	0	1	1

Necessary hardware:



Software implementation:

Use C programming language.

Operation 5:

Using the two signals produced by the optical sensors in the DC motor board, the objective is to calculate and display the motor's speed in rpm.

Suggestion: The signal produced by the optical sensor can be treated as an Input Capture interrupt of the **Timer1 (ICP1)** if connected to the **PD4**. One of the possible ways to calculate the motor's speed is to measure the time between two pulses generated by the optical sensor and calculating the respective conversion to rpm value.

A new operating mode must be added (Speed mode "V")

Character	Description
"S" or "s"	Switches Mode
"D" or "d"	Digital Mode
"A" or "a"	Analog Mode
"M" or "m"	Stepper Motor Mode
"V" or "v"	Speed Mode

Software implementation:

Use **C** programming language.